Overestimated Environmental Volatility Underpins Aberrant Decision Making in Type-I Bipolar Disorder

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Introduction

- Bipolar Disorder (BD) is a severe mood disorder characterized by recurrent mood episodes of mania and depression.
- Recent studies suggest that mood instability in BD may arise from altered reward perception and learning during decision making [1], [2].
- One key component of decision making is the inference of **environmental volatility** — how rapidly the environment changes.
- However, Only one study [3] has investigated volatility inference in BD, and it examined only a simple environmental structure in euthymic individuals.

Aim & Research Questions

Here, we tested whether individuals with BD show altered volatility inference across diverse environmental structures and whether it is associated with clinical symptoms.

RQ 1) Do individuals with BD show differences in behavioral patterns?
RQ 2) Do individuals with BD show altered volatility inference?
RQ 3) How do clinical symptoms relate to individual differences in volatility inference?

Method

Participants

HC (N=40) Type-I BD (N=32)

Measures

Mood
- Mania (YMRS)
- Depression (MADRS, BDI)

Trait
Hypomanic Personality Trait

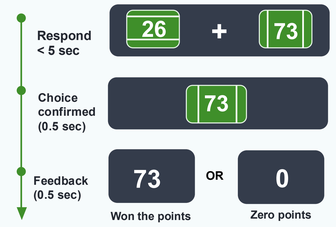
Motivational system
- BIS/BAS

Impulsiveness
- BIS-11

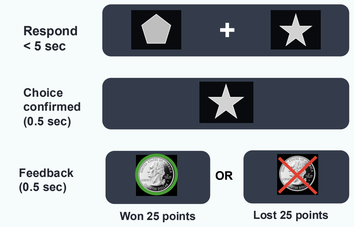
Cognitive function
- IQ
- Working memory

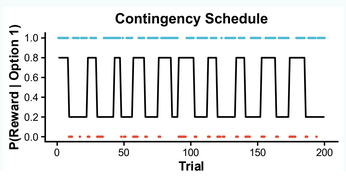
Task

PRL (Probabilistic Reversal Task) [4]

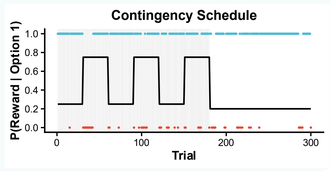


PTT (Probability Tracking Task) [5]



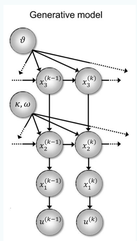


Different Pattern & Source of Environmental Change



Computational Modeling

Hierarchical Gaussian Filter [6]



Level	State of the world	Variable	Possible values	Generative model
3	Log-volatility of tendency	x_3	Real number	Gaussian random walk with constant step size ϑ
2	Tendency towards category "1"	x_2	Real number	Gaussian random walk with step size $\exp(\kappa x_3+\omega)$
1	Stimulus category	x_1	"1", "0"	Sigmoid trans-formation of x_2

Parameter of interest

μ_3^0 Prior Belief about Environmental Volatility

ζ Behavioral Stochasticity

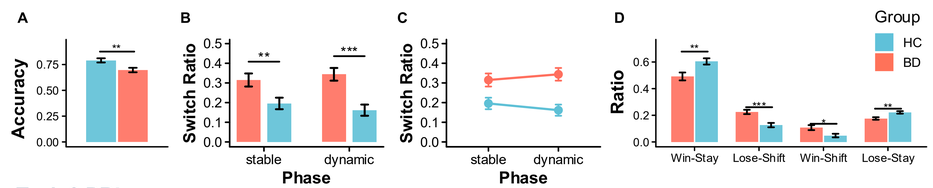
- Infer trial-wise hierarchical belief of the participant from behavioral data.

Hierarchical Bayesian analysis Estimate and compare group level parameter [7].

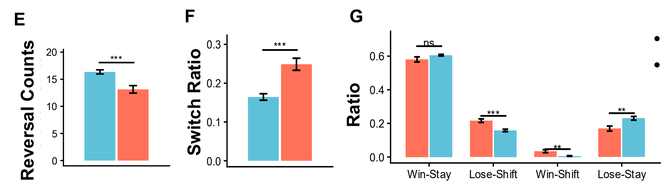
Results

Result 1. Group Differences in Behavioral Patterns

Task 1 PTT



Task 2 PRL

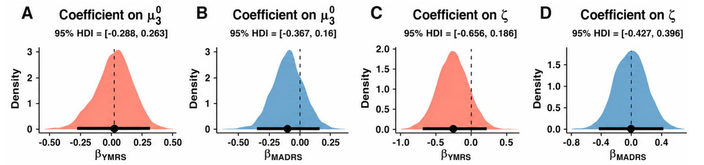


Note. p<.05 *, p<.01 **, p<.001 ***, Error bars indicate SEM.

Fig 1. The BD showed :
- **Poor performance** (A&E)
- **Higher option switching** (B&F) pronounced in dynamic phase (C), independent of previous outcomes (win vs. lose) (D&G).

Result 3. Association with Clinical Variables

Task 1 PTT



Task 2 PRL

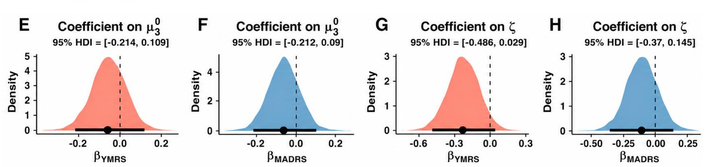
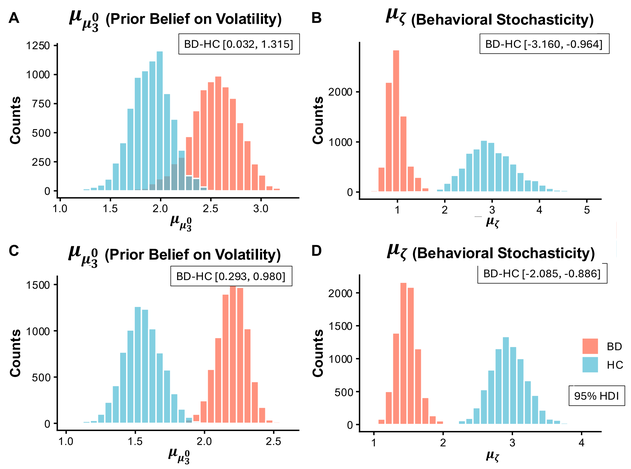
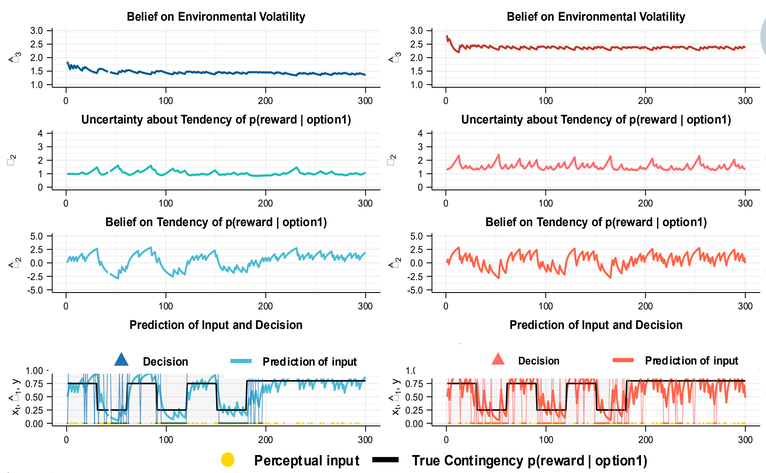


Fig 4. The mania (YMRS) and depression symptoms (MADRS) were not associated with prior volatility and behavioral stochasticity within BD.

Result 2. Group Differences in Perceptual Parameter





Higher volatility estimates across trials in BD

Greater overall higher uncertainty (∝ learning rates)

More frequent belief fluctuations

Higher switch rates at behavioral level

Fig 2. Group posterior distributions of HGF parameters of PTT (A&B) and PRL (C&D). The BD **exhibited higher priors on volatility** and higher choice randomness, which may underlie increased switching behavior.
Note. 95% Higher Density Intervals of posterior group differences are noted.

Fig 3. Belief trajectories and decisions of two representative participants whose posterior means are close to the group posterior means from each group HC (left); BD(right), in PTT.

Conclusion

Key Takeaway Type-I BD may be characterized by heightened volatility beliefs and increased behavioral stochasticity.

Clinical Implications

Explaining Clinical Symptoms Heightened volatility beliefs may lead individuals to interpret ordinary life events as signals of environmental change, potentially explaining why specific life events (i.e., personal achievements) triggers mood episodes.

Diagnostic Implication Volatility parameter was independent of depression severity – May serve as a diagnostic marker for differentiating BD from MDD.

Remaining Questions

Dependence on Episode Is altered volatility inference varies across episodes?
Origin of Altered Volatility Inference Does it precede illness onset vs. develop across episodes?
Predictive Utility Can volatility parameter improve prediction of heterogenous mood trajectories?

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